Seungdong Yoa

Contact Information

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Research Interests

Agentic AI & Multi-Agent Reasoning
Self-Evolving Agents & Foundation Models
Efficient AI & LLM Quantization
Anomaly Detection & LLM Reliability (Hallucination Detection)

Education

Korea University, Seoul, Korea
M.S. in Computer Science and Engineering — Sep 2019 – Feb 2022
GPA: 4.02 / 4.5 — Advisor: Hyunwoo J. Kim, Ph.D

Korea University, Seoul, Korea
B.S. in Computer Science and Engineering (Major) — Mar 2010 – Feb 2017
B.A. in Business Administration (Minor)
The National Scholarship for Science and Engineering (Full Merit-based)

Publications

1. **Seungdong Yoa**, Ye Seul Sim, Suhee Yoon, Sanghyu Yoon, Dongmin Kim, Soonyoung Lee, Junhyun Lee, Bumsoo Kim. ASQ: Agent-guided Semantic-aware Quantization for Large Language Models. *Advances in Neural Information Processing Systems* **(NeurIPS)**, 2026.
2. **Seungdong Yoa**, Ye Seul Sim, Suhee Yoon, Sanghyu Yoon, Dongmin Kim, Soonyoung Lee, Bumsoo Kim, Junhyun Lee. Efficient Multi-Agent Reasoning via Confidence-Guided Adaptive Debate. *International Conference on Machine Learning* **(ICML)**, 2026.
3. **Seungdong Yoa**, Sanghyu Yoon, Suhee Yoon, Dongmin Kim, Ye Seul Sim, Junhyun Lee, Woohyung Lim. From Static Benchmarks to Dynamic Protocol: Agent-Centric Text Anomaly Detection for Evaluating LLM Reasoning. *International Conference on Learning Representations* **(ICLR)**, 2026.
4. Sanghyu Yoon, Dongmin Kim, Suhee Yoon, Ye Seul Sim, **Seungdong Yoa**, Hyeseung Cho, Soonyoung Lee, Hankook Lee, Woohyung Lim. ReTabAD: A Benchmark for Restoring Semantic Context in Tabular Anomaly Detection. *International Conference on Learning Representations* **(ICLR)**, 2026.
5. Suhee Yoon, Sanghyu Yoon, Ye Seul Sim, **Seungdong Yoa**, Dongmin Kim, Soonyoung Lee, Hankook Lee, Woohyung Lim. AutoAnoEval: Semantic-Aware Model Selection via Tree-Guided LLM Reasoning for Tabular Anomaly Detection. *Findings of the Association for Computational Linguistics: EACL* **(EACL Findings)**, 2026.
6. **Seungdong Yoa**, Seungjun Lee, Hyeseung Cho, Bumsoo Kim, Woohyung Lim. ImagePiece: Content-aware Re-tokenization for Efficient Image Recognition. *AAAI Conference on Artificial Intelligence* **(AAAI)**, 2025.
7. **Seungdong Yoa**, Seungjun Lee, Chiyoon Kim, Hyunwoo J. Kim. Self-Supervised Learning for Anomaly Detection With Dynamic Local Augmentation. *IEEE Access*, 2021.

8. **Seungdong Yoa**, Jinyoung Park, Hyunwoo J. Kim. Learning Non-Parametric Surrogate Losses With Correlated Gradients. *IEEE Access*, 2021.
9. **Seungdong Yoa**, Minkyu Jeon, Youngjin Oh, Hyunwoo J. Kim. Learning to Balance Local Losses via Meta-Learning. *IEEE Access*, 2021.

Employment & Experience

LG AI Research, Seoul, Korea 2022 – Present
AI Research Scientist

- **Recursive Self-Quantization (RSQ):** Exploring failure-driven, training-free behavioral adaptation through self-evolving calibration environments and adaptive precision allocation.
- **Agent-driven Dynamic Workflows (AnoX):** Developed an AI agent platform orchestrating anomaly detection workflows dynamically using the Model Context Protocol (MCP), decoupling agents from tools for zero-code extensibility.
- **LLM Reasoning & Multi-Agent Systems:** Designed ATAD, a multi-agent benchmark protocol for evaluating LLM reasoning in text anomaly detection **(ICLR 2026)**.
- **LLM Reliability & Hallucination Detection:** Addressed numerical hallucination in Business Intelligence using agent-based token efficiency and a semantic-aware validation system.
- **Efficient LLM Reasoning (LASE):** Proposed a leader-centric multi-agent debate framework triggering interaction only in non-neutral regimes. Achieved multi-agent-level reasoning accuracy at near single-agent token cost **(ICML 2026)**.
- **Semantic-aware LLM Quantization (ASQ):** Engineered an agent-guided INT4 PTQ framework. Preserved domain-critical knowledge, matching or surpassing the zero-shot accuracy of dense FP16 models with zero inference overhead **(NeurIPS 2026)**.
- **Efficient Vision Models:** Authored "ImagePiece," a re-tokenization framework achieving SOTA efficiency in ViTs (1.5× faster, +0.39% accuracy), topping lightweight leaderboards **(AAAI 2025)**.
- **Industrial AI Optimization:** Deployed the 'Hero LensAA' project by applying Tab-Transformer and domain-specific regression adjustments, reducing overkill from 99% to near 0% on test sets. Awarded the **2024 LG Award**.

Hyundai Motor Group AIRS Company, Seoul, Korea July 2021 – Sep 2021
AI Researcher Intern

- Developed Anomaly Detection algorithms for Smart Factory quality control via self-supervised learning.

Korea University (M.S.), Seoul, Korea Sep 2019 – Feb 2022
Graduate Research Assistant

- Led a joint AI project with Samsung Electronics, developing real-time human-object interaction and anomaly detection algorithms for industrial drone monitoring systems.

Shinsegae E-commerce SSG.COM, Seoul, Korea Feb 2017 – Jun 2019
Software Engineer

- Engineered an OAuth 2.0 server and a scalable integration platform for AI speakers (Google Home, Naver Clova), and developed a customer service chatbot (CS Talk).

Entrepreneurial Experience

InfiniTensor Inc., Seoul, Korea — 2024 – Present
Founder & CEO

- **AgentMon:** Built and launched an autonomous marketing agent that receives commands and schedules through Discord and persistently executes content-posting workflows on X (Twitter), combining agent orchestration, tool use, and scheduled execution.
- **EffectGen / EffectGen HomeLiving:** Built generative-AI products for e-commerce businesses that transform product-only images into on-model fashion imagery and contextual home-living scenes for commercial product pages.

Honors and Awards

LG Award (2024): The most prestigious honor across all LG Group affiliates. Recognized for major contributions and successful commercial deployment in the Hero LensAA project.
AI Grand Challenge 2nd Place (Jan 2020): Awarded **$0.6M USD** research grant for AI problem-solving excellence.
National Scholarship for Science and Engineering (Full Merit-based, Top-tier academic performance), Korea

Extracurricular Activities

Future Entrepreneurs' Society (FES) — Sep 2014 – Dec 2015
Leading Member / HR Manager

- Designed manufacturing startup business models and led business strategy seminars in collaboration with top tech companies (Google Korea, Kakao).

Student Council — Mar 2010 – Dec 2011
General / Planning Manager

- Organized large-scale student activities and managed the council budget.

Patent

Wet type vacuum cleaner
- Registration number/date: 1010156040000 / 2011.02.10

Wet type air cleaner
- Registration number/date: 1009762030000 / 2010.08.10

Additional Information

Skills: Python, PyTorch, SQL, Next.js, Java, Spring Framework, LaTeX.

Clubs & Communities: FES Business Strategy Club, Korea Student Aid Foundation (KOSAF), KUBA, Student Council, Soccer Club.